

MSK Lower Extremity Physical Exam

Diagnostic Medicine III 2025

STARTING THE PATIENT ENCOUNTER

Introduces self to patient using first and last name and role.

General Survey Verbalizes General Survey while observing the patient

Vital Signs

- Takes or acknowledges abnormal blood pressure, heart rate, respiratory rate, temperature, weight,
- Set the agenda / Outline the events of the visit
- Wash hands/use sanitizer upon starting encounter
- Dons appropriate PPE

HIP

Inspect
(compare bilaterally)
(verbalize)
SSSDDM

- Symmetry
- Deformity same as UE
- Swelling same as UE
- Discoloration
- Scars
- Muscle atrophy same as UE

Palpate
(compare bilaterally)

- Anterior hip (iliac crest, iliac tubercle, anterior-superior iliac spine, pubic tubercle)
- Posterior hip (posterior-superior iliac spine, ischial tuberosity, sacroiliac joint)
- Greater trochanter and trochanteric bursa
- Femur (entire length)
- Verbalize: Palpating for tenderness, warmth, crepitus "Bony joint tenderness, crepitus, warmth"

Assess ROM (active and passive) "Lay supine"
(compare bilaterally)

- Hip flexion /extension
- Hip abduction/adduction
- Hip internal rotation/External rotation

KNEE/FORELEG

Inspect
(compare bilaterally)
(verbalize)
SSSDDM

- Symmetry
- Deformity
- Swelling
- Discoloration
- Scars
- Muscle atrophy

Palpation
(compare bilaterally)

- Patella & patellar tendon 1.
- Tibiofemoral joint 4.
- Tibiofemoral joint line (medial & lateral) 5.
- Medial meniscus 6.
- Lateral meniscus 7.
- Medial collateral ligament (MCL) 8.
- Lateral collateral ligament (LCL) 9.
- Tibial tubercle 3.
- Prepatellar and anserine bursa 2.
- Popliteal fossa 10.
- Foreleg 11.



Verbalize: Palpating for tenderness, warmth, bogginess, crepitus, enlargement, thickening

"Bony, joint tenderness, crepitus, warmth, bogginess, thickening, enlargement"

Assess ROM (active and passive) (compare bilaterally)	Flexion
	Extension
	Internal Rotation → knee and hip flexed at 90°, hold popliteal fossa in one hand and heel with other. rotate heel like you are waving w/ foot
	External Rotation
ANKLE/FOOT	
Inspect (compare bilaterally) <i>(verbalize)</i> SSSDDCC	Symmetry
	Deformity
	Swelling
	Discoloration
	Scars
	Calluses, corns or nodules
Palpation (compare bilaterally)	Lateral malleolus
	Lateral ankle ligaments
	Medial malleolus
	Medial ankle ligaments
	Ankle joint
	Calcaneus
	Achilles Tendon
	Gastrocnemius
	Soleus
	Plantar fascia
	Metatarsals
	Phalanges
	MTP, PIP, DIP joints
Verbalize: Palpating for tenderness, warmth, swelling, bogginess, crepitus	
Assess ROM (active and passive) (compare bilaterally)	Flexion (plantar flexion)
	Extension (dorsiflexion)
	Inversion
	Eversion
GAIT	
Inspect <i>(verbalize)</i> → Test for Trendelenburg after b/c pt standing	Base width (normal: 2-4 inches from heel to heel)
	Pelvic alignment
	Knee flexion
	Stride length
	Stance phase
	Swing phase
LE SPECIAL TESTS	
Lower limb measurement Also in MSK Spine Exam	Measured from the anterior iliac spine to the medial malleolus crossing the knee on the medial side. Perform bilaterally.
FABER (Flexion, ABduction, External Rotation) test (groin strain) "make a 4"	With the patient supine, position the leg into 90 degrees of flexion and externally rotate and abduct it so that the ipsilateral ankle rests distal to the knee of the contralateral leg. "Ankle sits on top of opposite thigh just above knee"
Trendelenburg Test (gluteal weakness)	Ask the patient to stand on one foot then shift their weight and stand on the other foot. Observe for a drop in the level of the iliac crests. Positive test - tilting of the pelvis on the contralateral side of the weakness. "The pelvis drops opposite to weak hip"



McMurray Test (meniscal tear)	With the patient supine, grasp the heel and flex the knee. Cup your other hand over the knee joint with fingers and thumb along the medial joint line. From the heel, externally rotate the lower leg, then push on the lateral side to apply a valgus stress on the medial side of the joint. At the same time, slowly extend the lower leg in external rotation. Repeat with internal rotation. Positive - Click felt or heard at joint line or tenderness at joint line
Lachman Test (ACL tear/laxity)	Patient is supine with knee flexed at 15-20 degrees. Wrap one hand around the distal femur to stabilize the femur. Place the other hand around the <u>proximal tibia</u> . While stabilizing the femur, pull the tibia anteriorly. ↳ posteriorly
Anterior Drawer Test (ACL tear/laxity)	Patient is supine. Place knee at 90 degrees flexion with foot flat on the table. Wrap both hands around proximal tibia with thumbs <u>placed on joint line</u> . Sit carefully on the patient's foot on the same side. Pull the tibia anteriorly and push the tibia posteriorly. Positive test - significant movement of the tibia. ↳ for Anterior Drawer Test ↳ for Posterior Drawer Test
Posterior Drawer Test (PCL tear/laxity)	Position the patient and place your hands in the positions described for the anterior drawer test. Sit on the patient's foot to minimize foot movement. Push the tibia posteriorly and observe the degree of backward movement in the femur.
Valgus Stress Test (MCL)	With the patient supine and the knee slightly flexed, move the thigh about 30° laterally to the side of the table. Place one hand against the lateral knee to stabilize the femur and the other hand around the medial ankle. Push medially against the knee and pull laterally at the ankle to open the knee joint on the medial side (valgus stress). Feel for excessive widening of the joint and lack of endpoint that may signal the ligament is no longer intact.
Varus Stress Test (LCL)	With the thigh and knee in the same position, change your position so that you can place one hand against the medial surface of the knee and the other around the lateral ankle. Push laterally against the knee and pull medially at the ankle to open the knee joint on the lateral side (varus stress). Feel for excessive widening of the joint and lack of endpoint that may signal the ligament is no longer intact.
Bulge Sign (Assess knee for minor effusion)	Patient is supine or has leg fully extended with support. Sweep your hand while applying pressure proximally along the medial side of the knee. Then apply pressure from proximal to distal on the lateral side of the knee. Observe for the bulge returning on the medial side.
Ballon Sign of Ballottement Test (Assess knee for major effusion)	Patient is supine with the leg fully extended. Place the thumb and index finger of your right hand on each side of the patella; with the left hand, compress the suprapatellar recess against the femur. Palpate for fluid ejected or "ballooning" into the spaces next to the patella under your right thumb and index finger. A palpable fluid wave is a Positive Test or a "Balloon Sign."
Thompson Test (Assess the integrity of the Achilles tendon)	Place the patient prone with the knee and ankle flexed at 90°, or, alternatively, ask the patient to kneel on a chair. Squeeze the calf and watch for plantar flexion at the ankle. Absent plantar flexion is a positive test for Achilles tendon rupture.
Consider whether a neurological examination is necessary when completing a musculoskeletal examination.	